

# AMBIENT SYSTEMS

Climate At Your Service

www.ambientsystems.ai

# Outline

1. Motivation
2. Problem and Market Trends
3. Solution and Technology
4. Applications
5. 
6. Ambient Benefits
7. Appendix

# Cyber-Physical Systems (CPS) are driving the next wave of Enterprise Applications

Machine Intelligence

Edge Devices

Cyber Security

Automation




Custom Applications

# There is increasing demand for system optimization beyond energy efficiency

## Problem

### 1. Limited Resources

It’s hard to find highly trained and experienced operators who can optimize multiple intelligent systems.

### 2. Technological Complexity

Many operate with outdated legacy systems and physical equipment that lack interoperability.

### 3. Competing Priorities

Different physical systems have different requirements based on the unique needs of occupants and operators.

# Market Drivers

## Productivity

1. Process Optimization
2. Enhanced Visibility
3. Faster Compliance

## Sustainability

1. Energy Efficiency
2. Carbon Management

## Cybersecurity

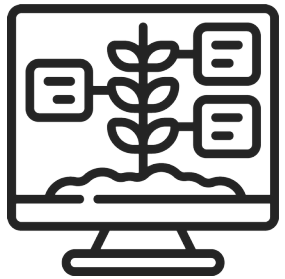
1. System Complexity
2. Increased Attack Surface

Ambient Systems is an enterprise software company that provides applied AI solutions for cyber-physical systems.

We have patent-pending invention that leverages AI to overcome challenges in the built environment.

These include AI- enabled planning, scheduling, and predictive maintenance, enhancing overall system performance, cybersecurity, and sustainability.

# Applications



**AgTech**

Indoor farming sustainability & optimization



**Healthcare**

Patient well-being



**Smart Homes**

Cyber security of smart home devices



**Retail**

Customer experience and inventory optimization

# Applied AI for Cyber-Physical Systems






## Sense

IoT gateway and sensors co-designed with manufacturing partners

## Optimize

NUS National University of Singapore

Patent Pending Invention from National University of Singapore (NUS)

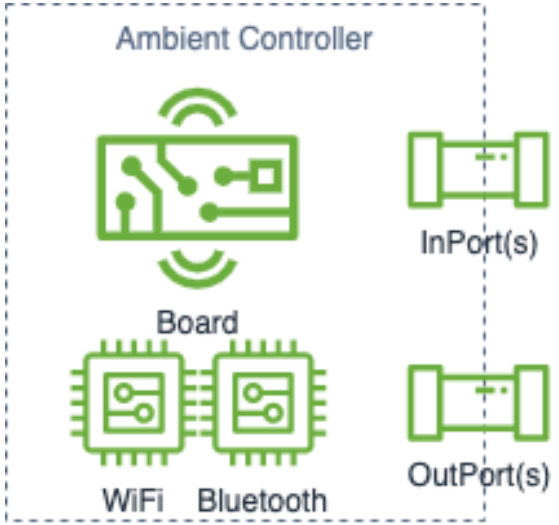


## Control

Offering end-to-end equipment controls after hardware development

# Technology

- IoT Gateway
  (Alternative Names: Hub, Terminal, Controller, Node)
  - Central hub for site-level equipment and networks
  - Communicates using various RFID frequencies and protocols
  - Ports for physical connection on site
  - Edge ML: device profiling and categorization
- User Application
  - Cloud-based web app (mobile to follow)
  - APIs for auxiliary systems
- Integrated Sensors
  - Integrated Air Sensors (Temperature, RH, CO2)
  - Additional Sensors Based on Use-Case




**Software Stack**

- Linux, C/C++
- Drivers: WiFi + Bluetooth
- Protocols: MQTT, ZigBee, LoRa, BacNET, Modbus
- Ports: Ethernet, RS-232

# What Makes Ambient Different

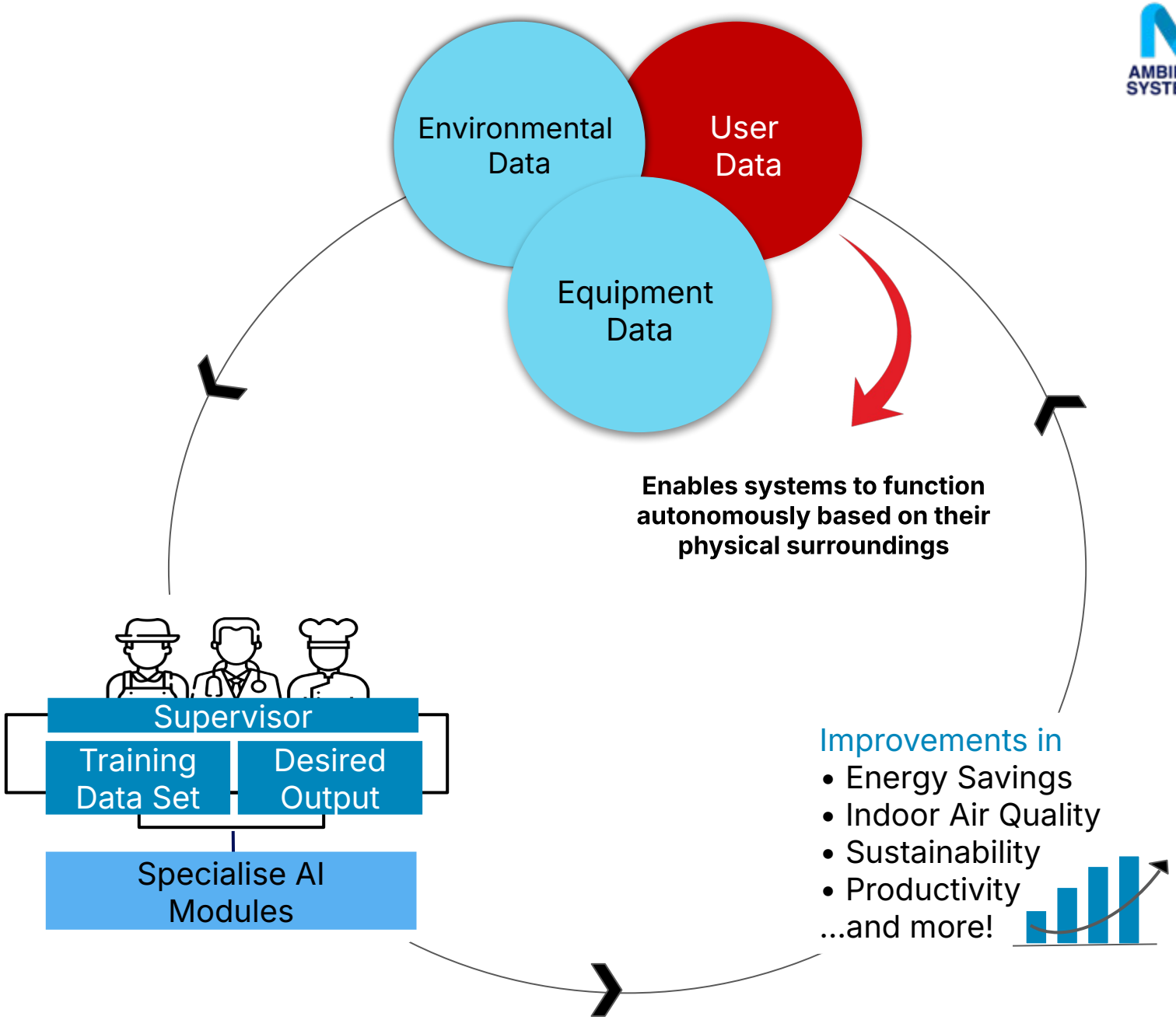
**Our Methodology:**
System optimization beyond energy efficiency.

**Our IP:**
Patent-pending invention for our launch market in AgTech.

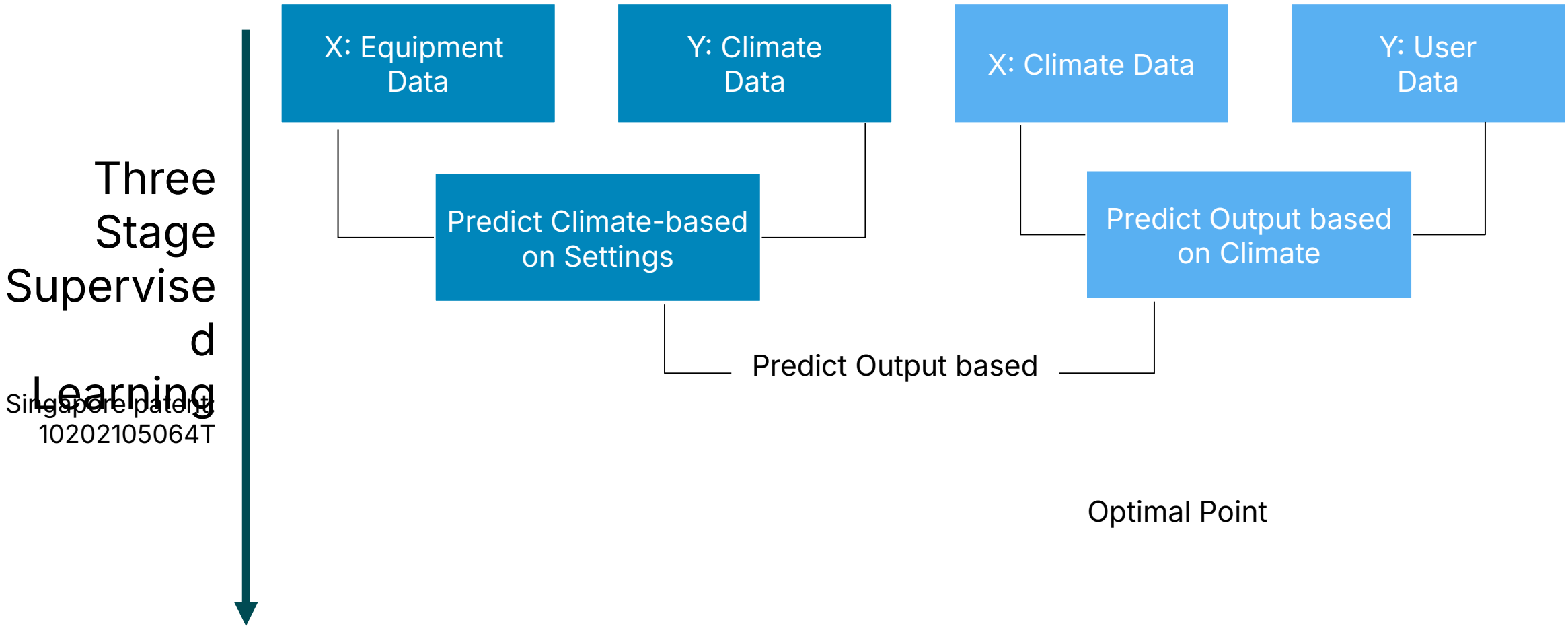
**Our Trade-secret:**
In-depth understanding of various industrial processes.

**Our Traction:**
University backing, 2 LOIs, pipeline of major companies.

# System Optimization beyond Energy Savings



# Underlying Magic: for Management of Ph



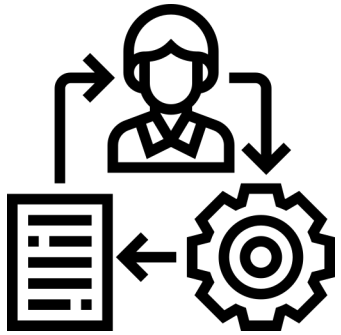

Singapore patent 10202105064T

https://www.techinnovation.com.sg/website/2 optimization-for-indoor-farming/

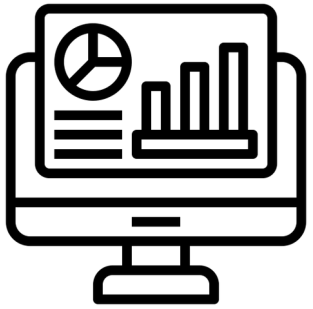
# Feature #1: Productivity



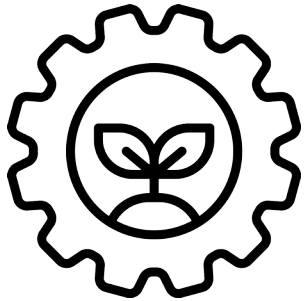
**1. Evaluate** the process needing improvement



**2. Embed** intelligent machines across the process

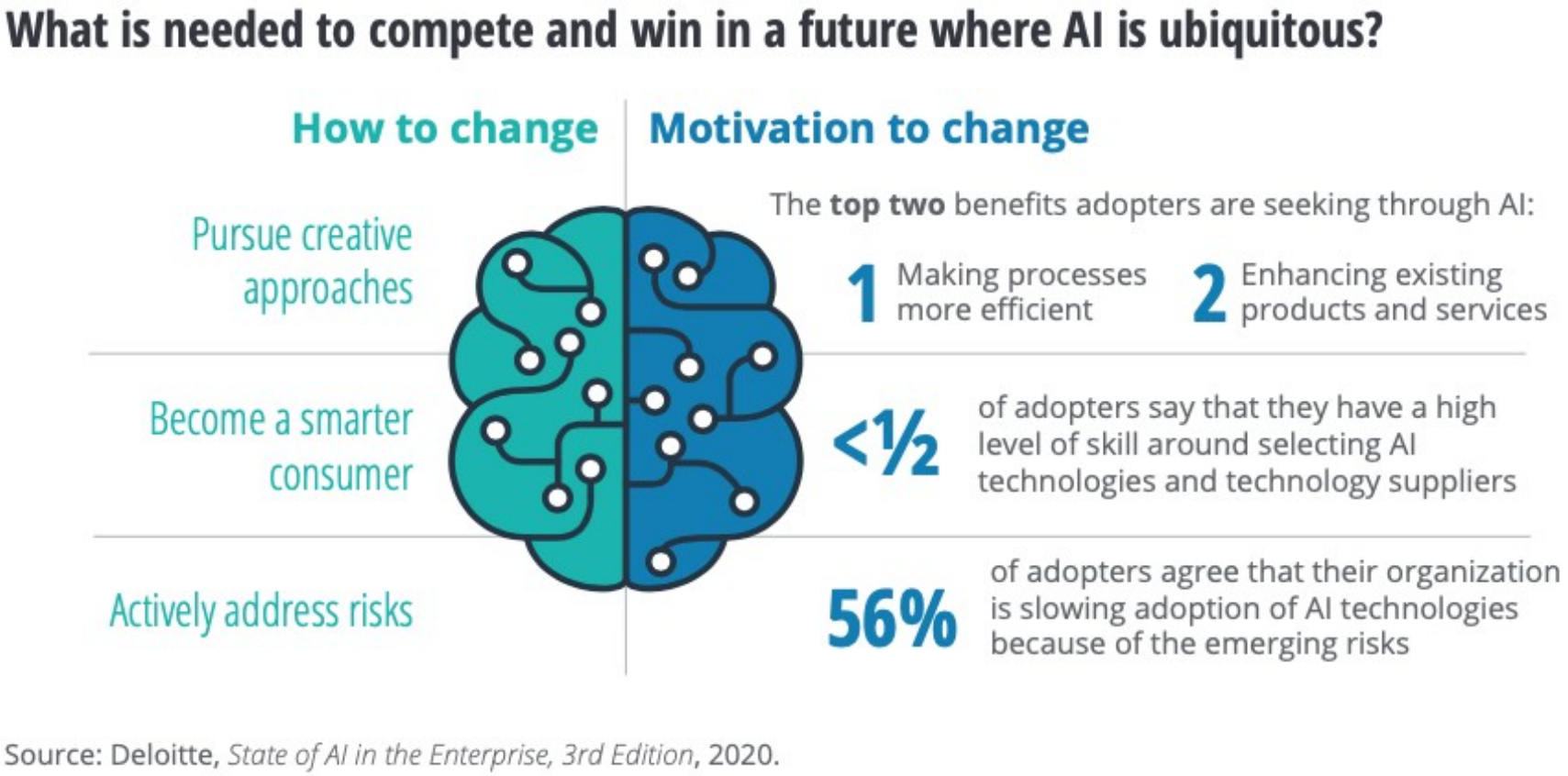


**3. Leverage** increased visibility, automated monitoring, compliance, and proprietary machine learning models from a single platform



**4. Launch** use-case: indoor farm sustainability and optimization.

# Market Trends: AI-enabled Productivity




Source: Deloitte

# Feature #2: Sustainability

1. Full visibility into certification requirements for sustainability.
2. Productivity needs balanced with energy efficiency.
3. Improved IAQ increased health and wellbeing of occupants, reducing healthcare costs and risks for respiratory illnesses.
4. Example: Indoor farms conditions and energy costs are optimized simultaneously.
5. Example: whole building ventilation is shifted to when electricity prices are lowest (e.g. night-time)



# Increasing Focus on Sustainability



**90%**

of consumers are concerned about the Environment

*Source: Sustainability Matters Now More Than Ever for Consumer Companies BCG, 2020*



**50%**

of CXOs see tackling climate change as their generation's top priority

*Source: Industry 4.0: At the intersection of readiness and responsibility Deloitte, 2020*



Ambitious targets to green the built environment in Singapore by 2030:

**80%** green buildings

**80%** improvements in energy efficiency

**80%** new buildings to be super lower energy

*Source: Industry 4.0: At the intersection of readiness and responsibility Deloitte, 2020*

Source: SP Digital

# Feature #3: Cyber Security

1. Site-level scripts use the device data stream to scan for anomalies.
2. Vulnerabilities are discovered and relayed to the user.
3. Leverages ML techniques for classification and anomaly detection.
4. Back-end Services delivered securely through AWS cloud.



# Market Trends: Cyber Security



**175zb**

predicted collective sum of the world's data by 2025

**Seagate 2018**

**65%** of Organisations have a shortage of Cybersecurity staff



**(ISC)² 2019**

**59%** of Companies have experienced a data breach caused by a vendor or third party



**Ponemon Institute 2018**

**51%** of Cybersecurity professionals say their organisation is at risk due to Cybersecurity staff shortage



**(ISC)² 2019**

Source: Deloitte

# Media and Links

- TechInnovation Tech Offer
- NUS GRIP Run 5 Post
- NUS GRIP Run 5 Media Post
- NUS GRIP Pitch
- LinkedIn: ambient-systems-ai
- Twitter: ambient_systems
- Crunchbase: ambient-systems
- Email: team@ambientsystems.ai

# Thank You

Visit us at: www.ambientsystems.ai

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# Appendix

# Increasing Energy Demand and Consumption




*Source: Our World in Data*

**In the built environment, it is imperative for buildings to be even more energy efficient and drive greater reduction in**

# Indoor Farming Overview

Improve crop yield in a market where only **51% of Indoor Farms are profitable**

**24% reduction in loss of crops** due to poor growing environment

Real-time improvements in farm operations efficiency resulting in **0.5 - 1.5 years** payback period



Indoor farm operators seeking to optimize crop performance through customized insights across key parameters.

# Healthcare Overview

Prevent **65-70%** of hospital-associated infections costing up to USD11B

Hospitals with superior patient experience achieve **50%** more in net margins

Improve air filtration and recirculation on hospital premise to prevent nosocomial outbreaks of COVID-19 **(Mortality rates >60%)**



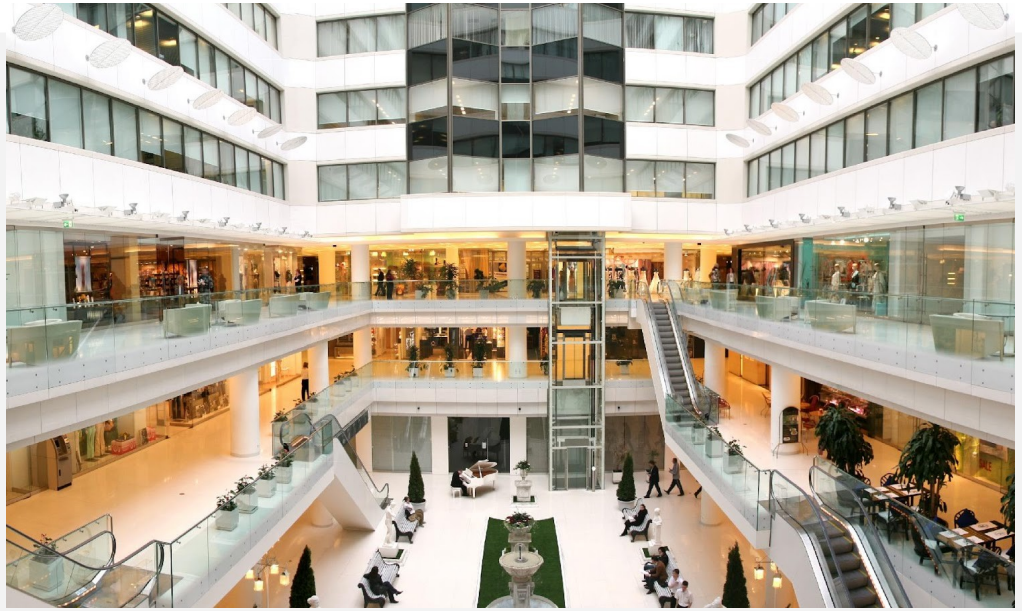
Hospitals looking to improve patient well-being hrough monitoring of indoor air quality

# Retail Overview

**70%** of customers avoid retail stores if they are too hot, too cold or too stuffy

Using retail heatmaps to optimise retail layout resulting in **5%** increase in sales

Improve in-store conversions by more than **10%** by analysing customer footfall trends



Retailers looking to optimize customer experiences through improving retail comfort and analysing footfall behavior

# Smart and Healthy Buildings Brings Value



## Smart office buildings can potentially garner a **37%**

rent premium per nett square feet, compared to buildings without smart characteristics

*Source: How Much Are Smart Buildings Really Worth? NAIOP Massachusetts, 2018*



## Healthy buildings increase productivity by **US$6,600 per employee**

translating to cost savings and increased revenue for tenants

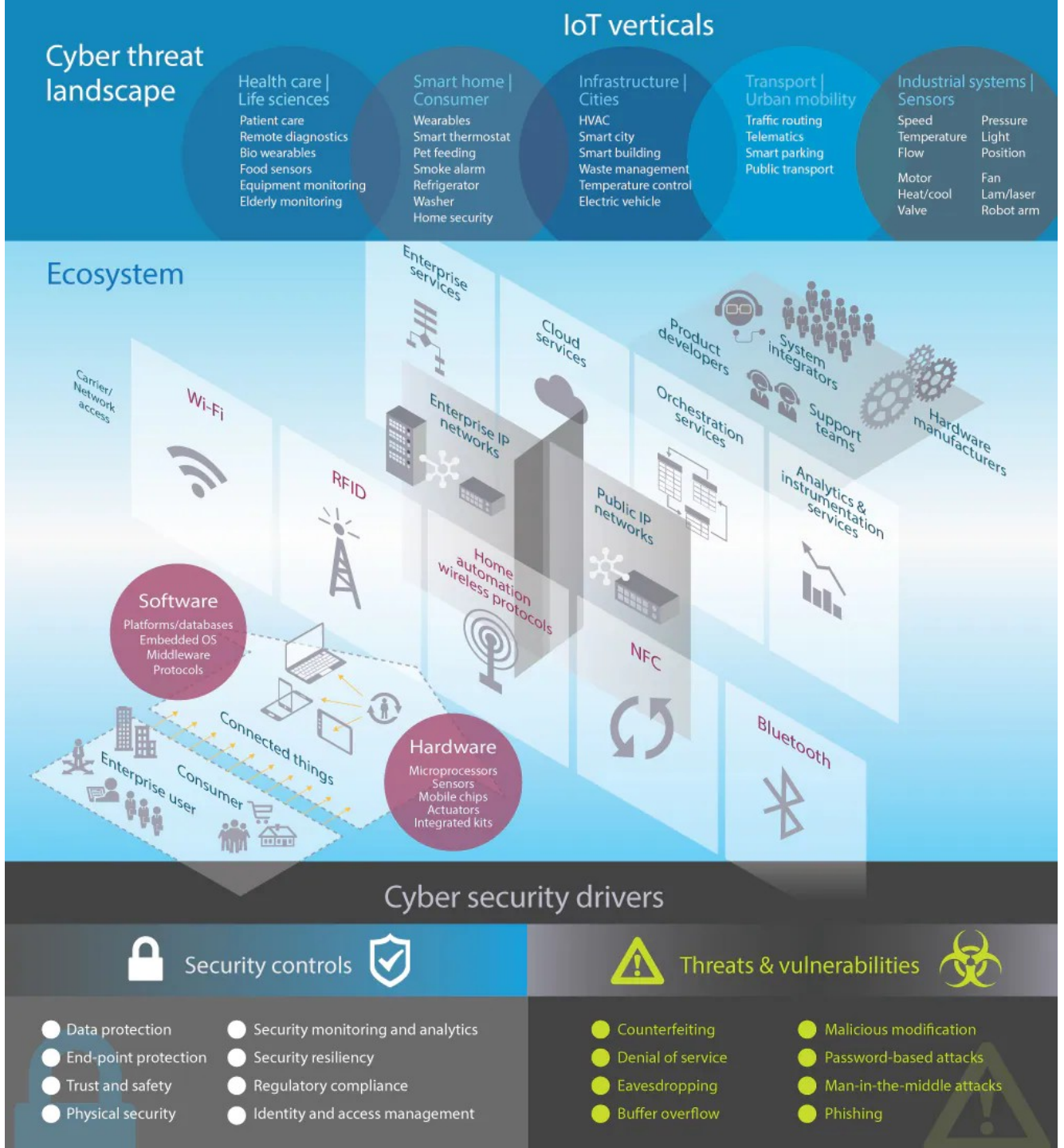
*Source: The Best Talent will Demand the Smartest Buildings Memoori Smart Building Research, 2018*



## Building analytics can reliably drive **3-9%**

energy savings annually, a figure expected to persist or even increase

*Source: Industry 4.0: At the intersection of readiness and responsibility Deloitte, 2020*




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# One step ahead with AI-enabled prescriptive maintenance

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| Artificial Intelligence | Machine Learning | Deep Learning |
|---|---|---|
| | | Data points gathered  |
| Automated Offerings<br>• Building Performance Optimisation<br>• Building Process Optimization | Smart Offerings<br>• Asset Optimization<br>• AI-based<br>• AI-based Lighting Control | Cognitive Offerings<br>• Predictive Maintenance |

Identify general issues ✓

Understand energy consumption by analysing primary datapoints ✓

Personalised control ✓

Understand energy flow and building occupancy ✓

Collect external datapoints ✓